

Bible of Economics of Technology and Management

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Lecture of 24/02/2026: introduction

1.1 Random concepts captured

Sustainability

The modern concept of sustainability was formalized in the 1980s in Northern Europe and politically promoted by Gro Harlem Brundtland, Prime Minister of Norway.

Definition 1.1 (Sustainable Development). *Development that meets the needs of the present without compromising the ability of future generations to meet their own needs.*

Sustainability is based on three interconnected pillars:

- Environmental protection
- Economic growth
- Social improvement

Taxes

Definition 1.2 (Tax). *Compulsory payment imposed by the government that increases the final price paid by consumers beyond the original market price of a product or service.*

A tax creates a wedge between the price paid by consumers and the price received by producers.

Ministry of Made in Italy

Definition 1.3 (Ministry of Made in Italy). *Italian governmental institution responsible for promoting, protecting, and developing national production, industrial policy, and the international competitiveness of Italian firms.*

Greenwashing

Definition 1.4 (Greenwashing). *The practice of communicating or promoting environmentally sustainable actions or policies without actually implementing substantial or verifiable environmental improvements.*

It creates a misleading perception of environmental responsibility.

Competition

Definition 1.5 (Competition). *Strategic process through which firms seek to gain market advantage either by minimizing production costs or by differentiating their products from those of competitors.*

Cost minimization leads to price-based competition, while product differentiation leads to value-based competition.

Opportunity Cost of Capital

Definition 1.6 (Opportunity Cost of Capital). *The expected return that investors forgo by investing resources in a specific project instead of in the best alternative investment with comparable risk.*

It represents the minimum return required to justify the allocation of capital to a given investment.

Economies of Scale

Definition 1.7 (Economies of Scale). *They occur when the average cost of production decreases as the quantity of output increases.*

They arise because fixed costs are spread over a larger number of units and because larger scale production may improve efficiency and bargaining power.

Consider bottled water. The average price per liter is typically lower when purchasing a large pack (e.g., a case of water bottles) rather than a single small bottle. This happens because fixed costs such as packaging, transportation, and distribution are spread over a larger quantity of product. As a result, the cost per unit decreases when buying in bulk.

Equilibrium between Demand and Offer

Definition 1.8 (Market Equilibrium). *The condition in which the quantity demanded by consumers equals the quantity supplied by producers.*

Economics studies how prices adjust to balance demand and supply, leading to an equilibrium allocation of goods and resources.

Supply Risk

Definition 1.9 (Supply Risk). *The possibility that the production of a good is disrupted due to limited availability or restricted access to essential raw materials.*

For example, smartphones contain small but significant amounts of gold and other rare materials. If these resources are geographically concentrated (e.g., controlled or largely available in specific countries) and access becomes restricted, firms may be unable to produce the final product. Dependence on critical raw materials therefore represents a strategic production risk.

Customer Perceived Value

Definition 1.10 (Customer Perceived Value). *The subjective evaluation that a consumer forms about a product, which may differ from its objective or technical characteristics.*

Perception is influenced by branding, marketing, reputation, and personal expectations, and it often differs from the product's real intrinsic value.

Digital Product Passport

Definition 1.11 (Digital Product Passport (DPP)). *A structured digital record that contains standardized information about a product's origin, materials, production processes, environmental impact, and lifecycle.*

It enables transparency, traceability, and sustainability monitoring across the entire value chain, supporting circular economy practices.

Energy Impact of Digitalization

Observation 1.1. *Digitalization requires significant energy consumption, often supplied by non-renewable sources, particularly for data centers, network infrastructures, and digital devices.*

As a consequence, the expansion of digital technologies may increase carbon emissions unless energy is sourced from renewable alternatives.

Sustainable Development Goals (SDGs)

Definition 1.12 (Sustainable Development Goals (SDGs)). *A set of 17 global objectives adopted by the United Nations to promote sustainable development by 2030, integrating economic growth, social inclusion, and environmental protection.*

They provide a common framework for governments, firms, and institutions to align policies and strategies with sustainability principles.

Strategic and Speculative Investments

Definition 1.13 (Strategic Investment). *An allocation of capital aimed at generating long-term value, competitive advantage, or sustainable growth for a firm.*

Definition 1.14 (Speculative Investment). *An allocation of capital primarily aimed at obtaining short-term financial gains by exploiting expected price fluctuations.*

10-Point Scale

Definition 1.15 (10-Point Scale). *Survey evaluation method in which respondents assign a score from 1 to 10 to express their judgment or level of satisfaction.*

Compared to a 5-point scale, a 10-point scale provides greater granularity and allows for more differentiated evaluations. In a 1–10 scale, the intermediate value is 5.5, which represents the theoretical midpoint of the evaluation range.

Willingness to Pay (WTP)

Definition 1.16 (Willingness to Pay (WTP)). *The maximum amount of money that a consumer is willing to spend to obtain a good or service.*

It represents the monetary expression of the value that a consumer assigns to a product.

Open Dialogue

Definition 1.17 (Open Dialogue). *A structured interaction among a group of individuals aimed at fostering transparent communication, shared understanding, and collaborative problem-solving.*

It encourages the exchange of different perspectives in order to reach more informed and inclusive decisions.

Substitute and Complementary Goods

Definition 1.18 (Substitute Goods). *Products that can replace each other in satisfying the same need. An increase in the price of one good typically increases the demand for its substitute.*

Definition 1.19 (Complementary Goods). *Products that are used together. An increase in the price of one good typically reduces the demand for its complement.*

1.2 Introduction to Business

Definition 1.20 (Business). *An organization (or economic system) that uses resources to satisfy customers' needs by providing goods or services demanded in the market.*

Goods and services are exchanged either directly (barter) or, more commonly, for money.

Definition 1.21 (Investment). *The allocation of financial, physical, or human resources with the expectation of generating future returns.*

Every business requires:

- an initial investment;
- a sufficient customer base;
- a product or service that meets market demand.

Definition 1.22 (Profit). *The positive difference between total revenues and total costs:*

$$\text{Profit} = \text{Revenues} - \text{Costs}$$

A business is **economically sustainable** only if revenues exceed costs over time.

1.2.1 Business Process

A business can be analyzed as a **process that transforms inputs into outputs**:



Definition 1.23 (Inputs). *The resources used by a firm to produce goods or services.*

Definition 1.24 (Outputs). *Goods and services that satisfy identified customer needs and are typically sold in the market to generate revenues.*

Definition 1.25 (Business Process). *The transformation of inputs into outputs in order to satisfy customer needs and generate profit.*

Organizations first identify **customer needs** and then acquire and combine resources (also called **production factors**) to produce one or more outputs.



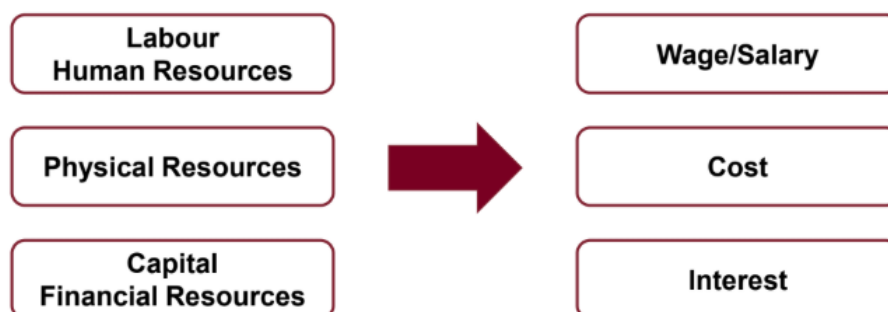
The main categories of inputs are:

- Labour (human resources)
- Physical resources
- Capital (financial resources)

The **production process** is usually distinguished into:

Definition 1.26 (Capital-Intensive Production). *A production process that relies primarily on physical and financial capital rather than human labour.*

Definition 1.27 (Labour-Intensive Production). *A production process that relies primarily on human labour rather than capital equipment.*



Each input used in production must be **supplied/purchased** and therefore **remunerated**.

Definition 1.28 (Remuneration of Inputs). *The payment made to resource owners in exchange for the use of production inputs.*

Typically:

- Labour (human resources) is remunerated through **wages/salaries**.
- Physical resources generate a **cost** (e.g., purchase, maintenance, depreciation).
- Capital (financial resources) is remunerated through **interest**.

For what concerns outputs, they have almost opposite characteristics depending on whether we are talking about products (goods) or services.

Definition 1.29 (Goods). *Tangible outputs that can be produced, stored, and transferred to consumers.*

Definition 1.30 (Services). *Intangible outputs produced and consumed through an interaction between provider and customer.*

The typical characteristics of goods are the following:

- **Tangibility:** goods can be inventoried and displayed; their features can be communicated.
- **Standardization:** goods can be produced with a predefined and replicable set of features.
- **Separation:** production, purchase, and usage are typically distinct phases.
- **Durability:** goods can often be stored, returned, or resold.

While the typical characteristics of services are the following:

- **Intangibility:** services cannot be inventoried or displayed like physical products.
- **Heterogeneity:** delivery and perceived quality vary across providers, customers, and contexts.
- **Inseparability:** production and consumption occur simultaneously; customers often participate in delivery.
- **Perishability:** services cannot be stored, returned, or resold.

1.2.2 Business Size

Organizations can be classified according to different criteria. One of the most commonly adopted classifications is based on **business size**, typically measured by the number of employees.

Definition 1.31 (Small and Medium Enterprises (SMEs)). *Firms with fewer than 250 employees.*

SMEs are usually divided into:

- **Micro Enterprises:** from 1 to 9 employees.
- **Small Enterprises:** from 10 to 49 employees.
- **Medium Enterprises:** from 50 to 249 employees.

Definition 1.32 (Large Enterprises). *Firms with 250 or more employees.*

1.2.3 Business Functions

Definition 1.33 (Business Functions). *Specialized organizational areas responsible for specific sets of activities necessary for the firm to operate and create value.*

Organizations, independently of their size, are typically structured into **six main functional areas**:



Marketing and Sales

- Defines products and services.
- Sets pricing strategies.
- Manages distribution channels (place).
- Develops promotion strategies.
- Oversees packaging.

Finance and Administration

- Ensures the availability of financial resources (capital).
- Manages accounting and administrative tasks.

Human Resources

- Recruitment and training of employees.
- Compensation and incentive systems.
- Job assignment and workforce planning.

Logistics and Operations Management

- Management of production processes.
- Quality control.
- Management of logistics flows and warehouses.

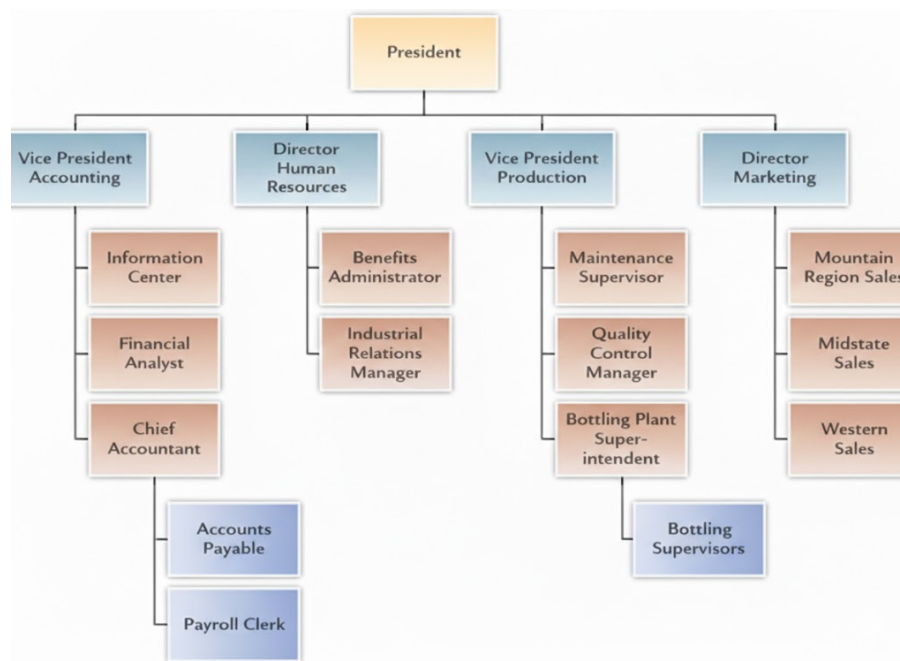
Research and Development (R&D)

- Research of new products/services.
- Development of new products/services.
- Improvement and upgrading of existing products.

Information Technology (IT)

- Management of IT infrastructures and digital systems.

1.2.4 Organizational Chart



Definition 1.34 (Organizational Chart). *A graphical representation of a company's internal structure, showing hierarchical relationships, roles, and lines of authority.*

In a typical hierarchical structure, authority flows from the top management (e.g., President) to functional managers (e.g., Accounting, Human Resources, Production, Marketing), and then to lower-level supervisors and operational staff.

This type of structure clarifies:

- who reports to whom;
- the division of responsibilities;
- the functional specialization within the organization.

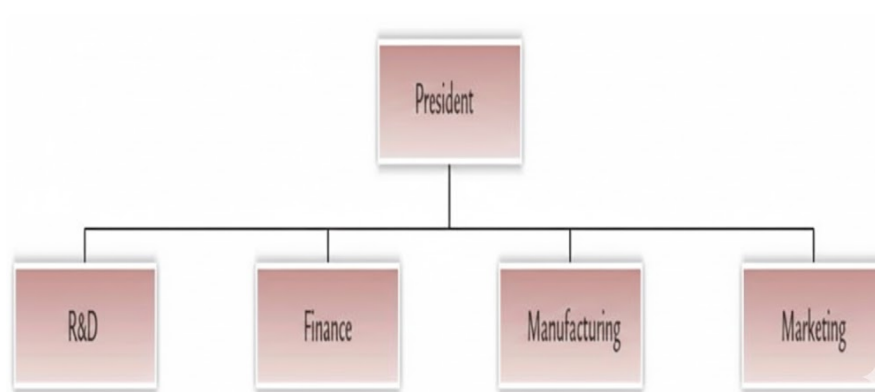
It is characteristic of a **functional organizational structure**, where activities are grouped by business functions.

1.2.5 Organizational Structure

Definition 1.35 (Organizational Structure). *It defines how tasks, responsibilities, authority, and communication are arranged within a firm.*

There are **three main types** of organizational structures.

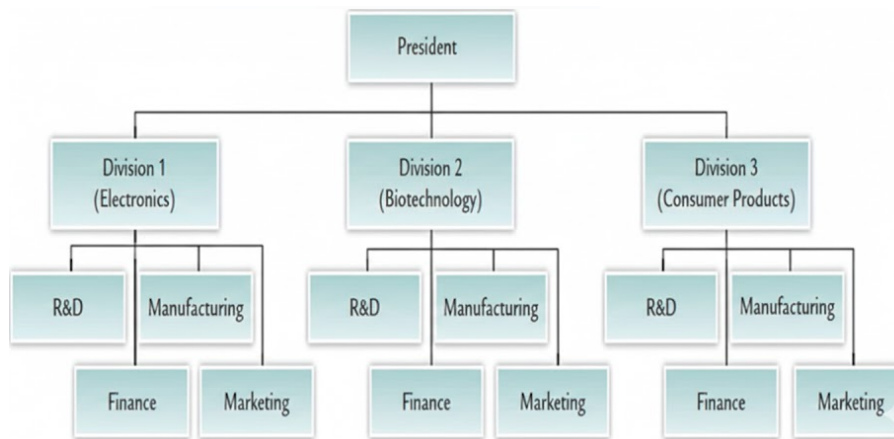
1.2.5.1 Functional Structure



Definition 1.36 (Functional Structure). *It groups activities according to business functions (e.g., R&D, Finance, Manufacturing, Marketing).*

This structure is generally the simplest organizational form. It centralizes activities by function and minimizes duplication of resources, leading to lower administrative costs. Since all departments report to top management, it promotes specialization and efficiency within each function but may reduce coordination across departments.

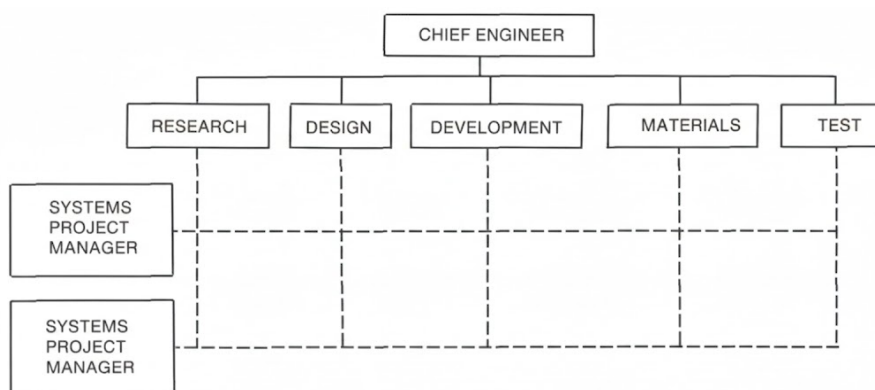
1.2.5.2 Divisional Structure



Definition 1.37 (Divisional Structure). *It organizes the company into semi-autonomous divisions based on products, markets, or geographic areas.*

Each division contains its own functional units (e.g., R&D, Manufacturing, Finance, Marketing). It increases flexibility and market focus but may duplicate resources across divisions, leading to higher costs. A divisional structure is typically adopted when a firm operates in highly differentiated products or heterogeneous markets that require distinct strategies, competencies, and managerial focus.

1.2.5.3 Matrix Structure



Definition 1.38 (Matrix Structure). *It combines functional and divisional structures, where employees report to both a functional manager and a project or product manager.*

This structure is more complex than both functional and divisional structures.

It introduces **bidirectional authority**, meaning that employees report simultaneously to:

- a functional manager (specialization dimension);

- a project/product manager (business or market dimension).

This structure combines high specialization with cross-functional coordination.

Definition 1.39 (Strong Matrix). *It gives greater authority to the project or product manager, while functional managers play a supporting role.*

Definition 1.40 (Weak Matrix). *It gives greater authority to functional managers, while project managers have limited coordination power.*

The matrix increases flexibility and integration but also raises coordination costs and potential conflicts due to dual reporting lines.

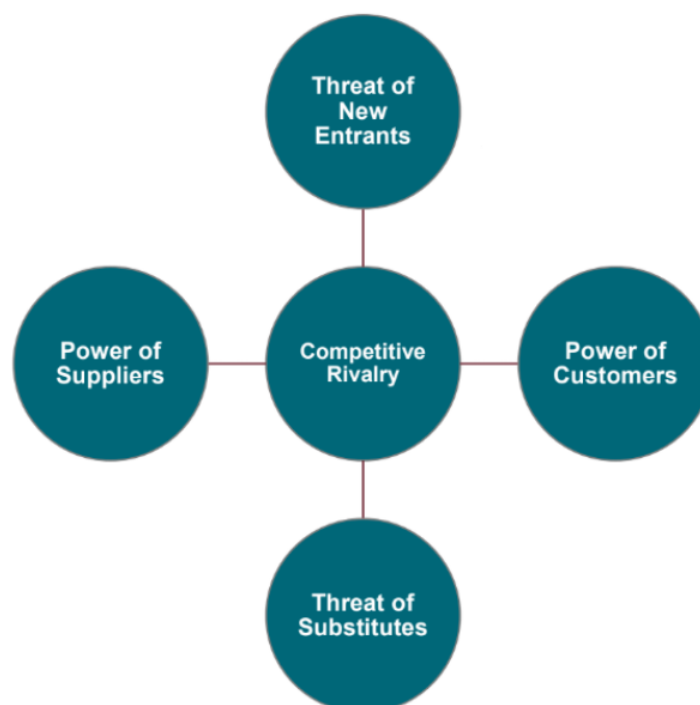
1.2.6 Competition

Definition 1.41 (Industry (sector)). *A group of firms operating in the same segment of the economy and sharing a similar type of business activity.*

Each industry includes one or more **markets**, where products and services are sold and where firms **compete** to achieve a favourable competitive position.

Definition 1.42 (Market). *The environment in which buyers and sellers interact to exchange goods or services.*

Porter's Five Forces Model (1985) is a framework used to analyze the competitive structure of an industry through **five key forces**.



- **Threat of New Entrants:** determined by entry barriers such as time, cost, specialized knowledge, and technology protection.
- **Power of Suppliers:** influenced by the number of suppliers, their size, uniqueness of inputs, and switching costs.
- **Power of Customers:** influenced by customer concentration, order size, price sensitivity, and switching costs.
- **Threat of Substitutes:** determined by the availability and performance of alternative products and switching costs.
- **Competitive Rivalry:** intensity of competition among existing firms, influenced by number of competitors, differentiation, and customer loyalty.

Definition 1.43 (Competitive Strategy). *The search for a favorable competitive position within an industry.*

Its objective is to achieve a profitable and sustainable position relative to competitors.

Definition 1.44 (Competitive Advantage). *An attribute, or a set of attributes, that enables a firm to outperform competitors and obtain a favorable competitive position.*

2

Lecture of 25/02/2026: Noreen_Ch01

2.1 Random concepts captured

Customer Engagement

Definition 2.1 (Customer Engagement). *The strategies and actions implemented by a firm to attract, involve, and retain customers by increasing their interaction and connection with the brand.*

Typical engagement tools include discounts, special offers, loyalty programs, and promotional campaigns aimed at stimulating customer participation and repeat purchases.

Work Capacity

Definition 2.2 (Work Capacity). *The amount of working time available to perform tasks within a given period.*

Before planning activities, it is necessary to verify the availability of working hours in order to allocate resources efficiently and avoid overload or underutilization.

Outsourcing

Definition 2.3 (Outsourcing). *The practice of contracting external firms or third parties to perform activities or services that were previously carried out internally.*

It is typically adopted to reduce costs, access specialized expertise, or increase organizational flexibility.

Strategy

Definition 2.4 (Strategy). *The long-term direction and scope of an organization, aimed at achieving competitive advantage and guiding future decisions.*

It defines how a firm allocates resources and positions itself in order to achieve its objectives over time.

Economics: Efficiency and Effectiveness

Definition 2.5 (Economics). *The study of how scarce resources are allocated, typically analyzed through quantitative measures and numerical evaluation.*

Economic analysis relies heavily on measurable variables such as costs, revenues, prices, and productivity.

Definition 2.6 (Efficiency). *The ability to achieve a given output using the minimum possible amount of resources.*

Definition 2.7 (Effectiveness). *The degree to which an objective or desired result is achieved.*

Efficiency concerns *how well* resources are used, while effectiveness concerns *whether* the intended goal is reached.

Financial vs. Economic Perspective

Definition 2.8 (Financial Perspective). *It concerns only monetary flows, particularly cash inflows and outflows (cash flow).*

It focuses on liquidity, revenues, costs, and the firm's ability to generate and manage money.

Definition 2.9 (Economic Perspective). *It concerns the creation of value through the efficient transformation of skills, resources, and services into outputs that generate utility and market value.*

While the financial dimension measures money movements, the economic dimension evaluates the firm's ability to create value from available resources.

Invention vs. Innovation

Definition 2.10 (Invention). *The creation of a new idea, concept, technology, or method.*

Definition 2.11 (Innovation). *The successful commercialization or practical application of an invention in the market.*

An invention becomes an innovation only when it generates economic value through market adoption.

Cost vs. Revenue

Definition 2.12 (Cost). *A monetary outflow (cash outflow) sustained by a firm for acquiring resources or conducting business activities.*

Definition 2.13 (Revenue). *A monetary inflow (cash inflow) generated from the sale of goods or services.*

Stakeholders vs. Shareholders

Definition 2.14 (Stakeholders). *All individuals or groups that have an interest in or are affected by the activities and performance of a firm.*

Stakeholders include shareholders, employees, customers, suppliers, creditors, governments, and the local community.

Definition 2.15 (Shareholders). *Individuals or entities that own shares of a company and therefore hold a financial ownership interest in it.*

All shareholders are stakeholders, but not all stakeholders are shareholders.

2.2 Managerial Accounting and Cost Concepts

Definition 2.16 (Financial Accounting). *Concerned with the preparation and reporting of financial information for external users, such as shareholders, creditors, regulators, and tax authorities.*

Its main objective is to provide standardized and reliable financial statements describing the economic performance and financial position of the firm.

Definition 2.17 (Managerial Accounting). *Concerned with providing financial and non-financial information to internal managers in order to support planning, control of operations, and decision-making.*

Unlike financial accounting, managerial accounting is internally oriented and not subject to strict external reporting standards.

Definition 2.18 (Commercial Activity). *It consists of purchasing finished goods and reselling them to customers without transforming them.*

The firm acts as an intermediary between producers and consumers.

Definition 2.19 (Manufacturing Activity). *It consists of purchasing raw materials or semi-finished goods, transforming them through a production process, and selling the resulting finished products.*

In this case, value is created through the transformation process.

2.2.1 Cost Classification

Costs are classified for five main purposes:

1. assigning costs to cost objects;
2. accounting for costs in manufacturing companies;
3. preparing financial statements;
4. predicting cost behavior in response to changes in activity;
5. supporting managerial decision-making.

The following cost classifications arise from **purpose 1**:

Definition 2.20 (Direct Cost). *A cost that can be easily and economically traced to a specific cost object.*

Definition 2.21 (Indirect Cost). *A cost that cannot be easily traced to a specific cost object and must be allocated.*

The following cost classifications arise from **purpose 2**:

Definition 2.22 (Manufacturing Costs). *Costs directly associated with the production process.*

They include:

- direct materials;
- direct labor;
- manufacturing overhead.

Definition 2.23 (Nonmanufacturing Costs). *Costs not directly related to production.*

They include:

- selling costs;
- administrative costs.

The following cost classifications arise from **purpose 3**:

Definition 2.24 (Product Costs (inventoriable costs)). *Costs assigned to products and included in inventory until the product is sold.*

Definition 2.25 (Period Costs). *Costs expensed in the period in which they are incurred.*

The following cost classifications arise from **purpose 4**:

Definition 2.26 (Variable Cost). *Cost that changes proportionally with the level of activity. A variable cost per unit is constant.*

Definition 2.27 (Fixed Cost). *Cost that remains constant in total within the relevant range of activity. If expressed on a per-unit basis, the average cost per unit varies inversely with the level of activity.*

Definition 2.28 (Mixed Cost). *Cost that contains both fixed and variable components.*

The following cost classifications arise from **purpose 5**:

Definition 2.29 (Differential Cost). *Cost that differs between alternative decisions.*

Definition 2.30 (Sunk Cost). *Cost that has already been incurred and cannot be recovered; it should not influence future decisions.*

Definition 2.31 (Opportunity Cost). *The benefit foregone by choosing one alternative over another.*

2.2.2 LO1¹: Assigning Costs to Cost Objects

Definition 2.32 (Cost Object). *Anything for which cost data are desired, such as a product, service, customer, project, or department.*

Costs assigned to cost objects can be classified as **direct** (2.20) or **indirect** (2.21).

Typical examples of direct costs include:

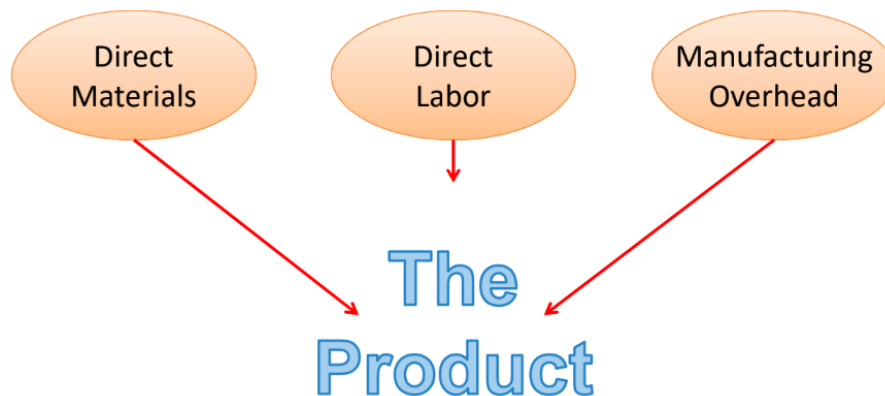
- direct materials;
- direct labor.

Typical examples of indirect costs include:

- manufacturing overhead;
- common costs.

Definition 2.33 (Common Cost). *Indirect cost incurred to support multiple cost objects and cannot be traced to any single cost object.*

2.2.3 LO2: Classifications of Manufacturing Costs



Manufacturing costs (2.22) are classified into three main categories:

- direct materials;
- direct labor;
- manufacturing overhead.

These three elements together form the **total manufacturing cost** of a product.

¹LO means Learning Objective.

Direct Materials and Direct Labor

Definition 2.34 (Direct Materials). *Raw materials that become an integral part of the finished product and can be conveniently traced to it.*

Example: A radio installed in an automobile.

Definition 2.35 (Direct Labor). *It consists of labor costs that can be easily and conveniently traced to individual units of product.*

Example: Wages paid to automobile assembly workers.

Manufacturing Overhead

Definition 2.36 (Manufacturing Overhead). *It includes all manufacturing costs except direct materials and direct labor. These costs cannot be directly traced to specific units of product.*

Manufacturing overhead includes:

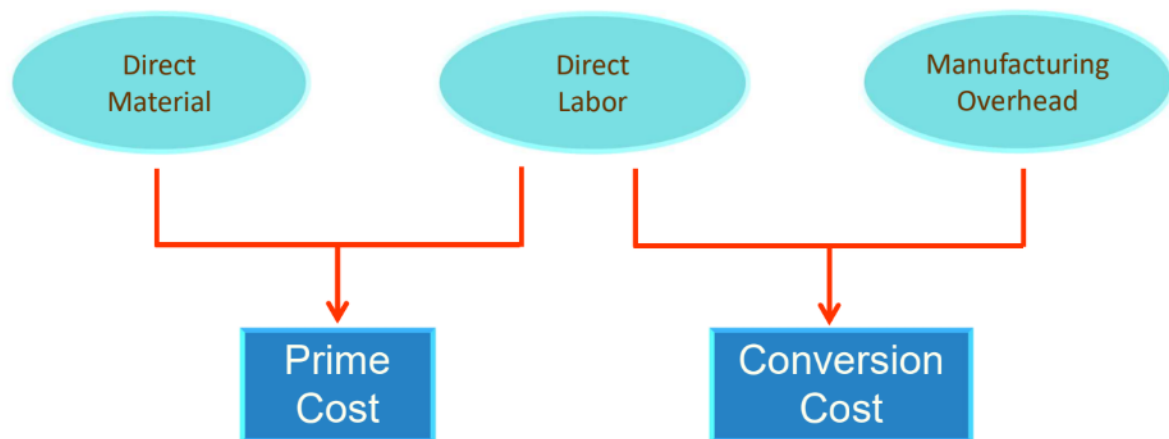
- indirect materials;
- indirect labor;
- other factory-related indirect costs.

Examples:

- lubricants and cleaning supplies used in the plant;
- maintenance workers and security guards;
- depreciation of manufacturing equipment;
- utility costs of the factory;
- property taxes and insurance on the manufacturing facility.

Only indirect costs associated with operating the factory are included in manufacturing overhead.

Prime and Conversion Costs



Definition 2.37 (Prime Cost). *The sum of direct materials and direct labor:*

$$\text{Prime Cost} = \text{Direct Materials} + \text{Direct Labor}.$$

Definition 2.38 (Conversion Cost). *The sum of direct labor and manufacturing overhead:*

$$\text{Conversion Cost} = \text{Direct Labor} + \text{Manufacturing Overhead}.$$

Nonmanufacturing Costs

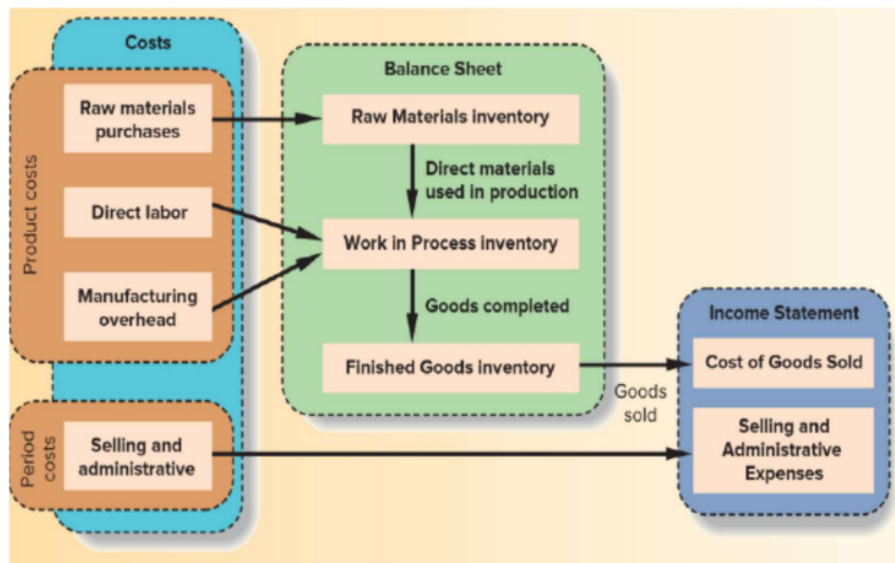
Nonmanufacturing costs (2.23) are classified into:

- selling costs (direct or indirect);
- administrative costs (direct or indirect);

Definition 2.39 (Selling Costs). *Costs incurred to secure customer orders and deliver the product.*

Definition 2.40 (Administrative Costs). *Costs that include executive, organizational, and clerical expenses necessary to manage the company.*

2.2.4 LO3: Preparing Financial Statements



Product costs (2.24) *attach* to units of product as they are purchased or manufactured and remain in inventory until the product is sold.

In manufacturing firms, product costs are composed of:

- raw material purchases (direct materials);
- direct labor;
- manufacturing overhead.

Period costs (2.25) typically include selling (2.39) and administrative costs (2.40).

Definition 2.41 (Balance Sheet). *Financial statement that reports a company's assets, liabilities, and owner's equity at a specific point in time.*

Definition 2.42 (Income Statement). *Financial statement that reports revenues, expenses, and net income over a specific period of time.*

While the balance sheet represents the financial position at a given date, the income statement measures performance over a time interval. Both are based on the **fundamental accounting equation**.

Definition 2.43 (Accounting Equation). *Equation that expresses the relationship between a firm's resources and the claims on those resources:*

$$\text{Assets} = \text{Liabilities} + \text{Equity}.$$

Definition 2.44 (Assets). *Economic resources owned or controlled by a company that are expected to provide future economic benefits.*

Definition 2.45 (Liabilities). *Present obligations of a company arising from past events, which require the future transfer of assets or services to other parties.*

Definition 2.46 (Equity (Owner's Equity)). *It represents the residual interest in the assets of a company after deducting liabilities.*

- Product costs are recorded as **inventory** on the **balance sheet** and become an expense only when the goods are sold (COGS) on the **income statement**.
- Period costs are recorded directly as **expenses** on the **income statement**.

Definition 2.47 (Raw Materials Inventory). *It includes materials that will be used in production and become part of the final product.*

Definition 2.48 (Work in Process (WIP) Inventory). *It consists of units that are partially complete and will require further processing before sale.*

Definition 2.49 (Finished Goods Inventory). *It consists of completed units that have not yet been sold to customers.*

In a manufacturing company, product costs flow through inventory accounts as production progresses:

- When direct materials are used in production, their costs move from **Raw Materials** to **Work in Process**.
- Direct labor and manufacturing overhead are added to **Work in Process** to convert materials into finished goods.
- When units are completed, costs are transferred from **Work in Process** to **Finished Goods**.
- When finished goods are sold, costs are transferred from **Finished Goods** to **Cost of Goods Sold**.

Definition 2.50 (Cost of Goods Sold (COGS)). *The cost of product units that have been sold during the period; it is recognized as an expense in the income statement.*

EXAMPLE (Balance Sheet)

Example Company Balance Sheet December 31, 2016			
ASSETS		LIABILITIES	
Current assets		Current liabilities	
Cash	\$ 2,100	Notes payable	\$ 5,000
Petty cash	100	Accounts payable	35,900
Temporary investments	10,000	Wages payable	8,500
Accounts receivable - net	40,500	Interest payable	2,000
Inventory	31,000	Taxes payable	6,000
Supplies	3,800	Warranty liability	1,100
Prepaid insurance	1,500	Unearned revenues	1,500
Total current assets	<u>89,000</u>	Total current liabilities	<u>61,000</u>
Investments	<u>36,000</u>	Long-term liabilities	
Property, plant & equipment		Notes payable	20,000
Land	5,500	Bonds payable	400,000
Land improvements	6,500	Total long-term liabilities	<u>420,000</u>
Buildings	180,000		
Equipment	201,000	Total liabilities	<u>481,000</u>
Less: accum depreciation	(56,000)		
Prop, plant & equip - net	<u>337,000</u>		
Intangible assets		STOCKHOLDERS' EQUITY	
Goodwill	105,000	Common stock	110,000
Trade names	200,000	Retained earnings	220,000
Total intangible assets	<u>305,000</u>	Accum other comprehensive income	9,000
Other assets	<u>3,000</u>	Less: Treasury stock	(50,000)
Total assets	<u>\$ 770,000</u>	Total stockholders' equity	<u>289,000</u>
		Total liabilities & stockholders' equity	<u>\$ 770,000</u>

The notes to the sample balance sheet have been omitted.

EXAMPLE (Income Statement)

Example Corporation Income Statement For the year ended December 31, 2016	
Sales (all on credit)	\$500,000
Cost of goods sold	<u>380,000</u>
Gross profit	<u>120,000</u>
Operating expenses	
Selling expenses	35,000
Administrative expenses	<u>45,000</u>
Total operating expenses	<u>80,000</u>
Operating income	40,000
Interest expense	<u>12,000</u>
Income before taxes	28,000
Income tax expense	<u>5,000</u>
Net income after taxes	<u>\$ 23,000</u>

2.2.5 LO4: Predicting Cost Behavior

Definition 2.51 (Cost Behavior). *The way in which a cost reacts to changes in the level of activity.*

The cost classifications involved in this concept are:

- variable costs (2.26);
- fixed costs (2.27);
- mixed costs (2.28).

Variable Costs

Definition 2.52 (Activity Base (Cost Driver)). *A measure of what causes the incurrence of a variable cost.*

Examples of activity bases:

- units produced;
- machine hours;
- labor hours;
- miles driven.

Fixed Costs

There are two types of fixed costs:

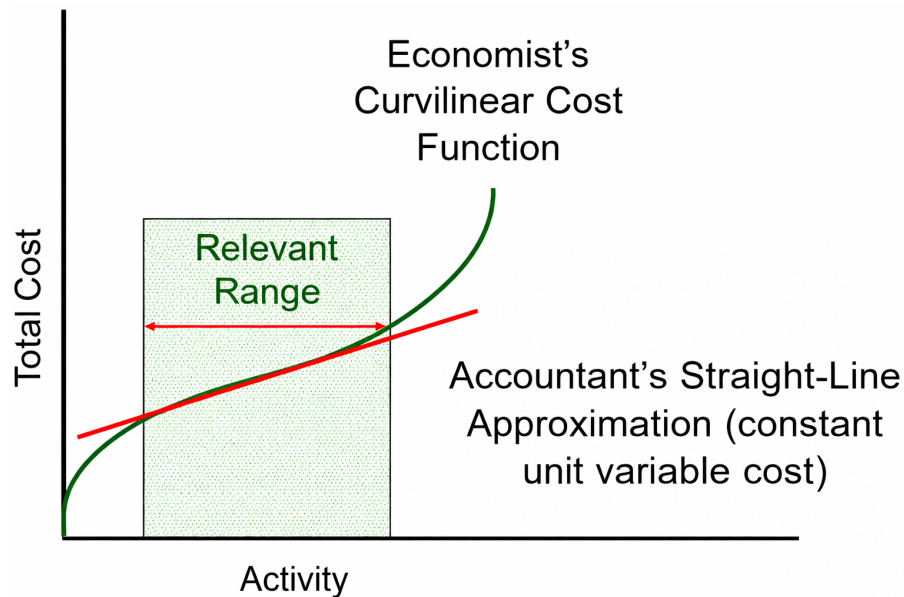
Definition 2.53 (Committed Fixed Cost). *Long-term fixed cost that cannot be significantly reduced in the short term.*

Definition 2.54 (Discretionary Fixed Cost). *Fixed cost that may be altered in the short term by current managerial decisions.*

Linearity Assumption and Relevant Range

Definition 2.55 (Relevant Range). *The range of activity within which cost behavior assumptions are valid.*

Observation 2.1 (Linearity Assumption). *Within the relevant range, total cost can be approximated by a linear function of the activity level.*

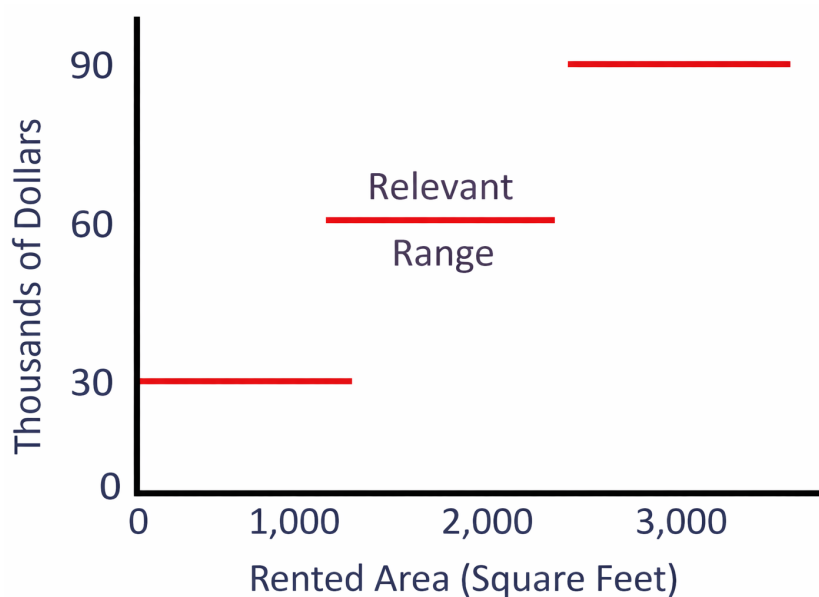


From a theoretical standpoint, total cost is typically **curvilinear**: as the level of activity increases, costs may initially increase at a decreasing rate and later at an increasing rate due to inefficiencies, congestion, overtime, or capacity constraints.

In managerial accounting, total cost is approximated using a **linear function**, assuming that:

- variable cost per unit is constant;
- total fixed cost remains constant.

Within the relevant range, the straight-line approximation closely follows the true (curvilinear) cost function. Outside the relevant range, the linear model may no longer provide an accurate representation of cost behavior.



Sometimes fixed costs increase in **steps** rather than continuously. For example, suppose office space costs \$30,000 per year for each additional 1,000 square feet rented. In this case, total fixed cost increases in a step fashion:

- From 0 to 1,000 sq ft → \$30,000
- From 1,001 to 2,000 sq ft → \$60,000
- From 2,001 to 3,000 sq ft → \$90,000

The cost is **fixed within each interval** (horizontal segment), but it increases in a **step pattern** when activity exceeds the current capacity. Therefore:

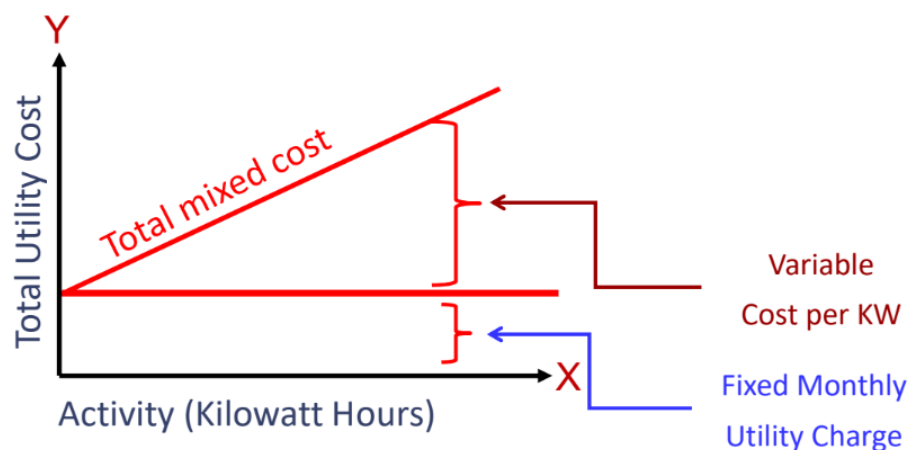
- within each step, the fixed cost is constant;
- when activity moves beyond the current capacity, total fixed cost jumps to a higher level.

The **relevant range** is the specific activity interval over which the fixed cost remains constant (i.e., where the graph is flat).

Comparison of Cost Behavior (Within the Relevant Range)

- **Variable Cost**
 - In total: changes proportionally with activity.
 - Per unit: remains constant.
- **Fixed Cost**
 - In total: remains constant.
 - Per unit: decreases as activity increases and increases as activity decreases.

Mixed Costs



A mixed cost has:

- A **fixed component** (incurred even if activity is zero).
- A **variable component** (changes proportionally with activity).

In the graph:

- The vertical intercept represents the **fixed cost**.
- The slope of the line represents the **variable cost per unit**.
- The total line represents the **total mixed cost**.

The total mixed cost can be expressed through a linear function:

$$Y = a + bX$$

Where:

- Y = total cost
- a = total fixed cost (vertical intercept)
- b = variable cost per unit (slope)
- X = level of activity

For example, suppose:

- Fixed monthly utility charge = \$40
- Variable cost per kilowatt hour = \$0.03
- Monthly activity level = 2,000 kilowatt hours

Then:

$$Y = 40 + (0.03 \times 2,000)$$

$$Y = 40 + 60$$

$$Y = \$100$$

Therefore, the total monthly utility bill is \$100.

2.3 Quick Checks

1. Which of the following costs would be considered a **period** rather than a **product** cost in a manufacturing company?

- A. Manufacturing equipment depreciation.
- B. Property taxes on corporate headquarters.
- C. Direct materials costs.
- D. Electrical costs to light the production facility.
- E. Sales commissions.

Correct answer(s): B, E.

2. Which of the following costs would be **variable** with respect to the number of ice cream cones sold at a Baskins & Robbins shop? (There may be more than one correct answer.)

- A. The cost of lighting the store.
- B. The wages of the store manager.
- C. The cost of ice cream.
- D. The cost of napkins for customers.

Correct answer(s): C, D.

3

Lecture of 03/03/2026: Noreen_Ch02

3.1 Random concepts captured

Economist Perspective

Observation 3.1. *An economist focuses on analyzing the **percentage composition of costs**. In particular, the objective is to identify which cost components represent the largest share of the total cost in order to better understand cost structure and economic behavior.*

Saturation Level

Definition 3.1 (Saturation Level). *The degree to which the available production capacity is utilized.*

When demand increases, firms may increase production by using machinery more intensively or by installing additional machines. In these situations, variable costs increase because more resources (such as labor, energy, and materials) are consumed.

Production capacity is not always fully usable. For example, machines may require maintenance, during which production must stop.

If a machine is theoretically available for 24 hours per day but is used for 18 hours, the saturation level corresponds to those 18 hours of effective utilization.

When demand exceeds the available production capacity, firms may also resort to **outsourcing**, delegating part of the production to external suppliers due to the lack of internal resources or machinery.

Relevance of The Information

Definition 3.2 (Relevant Information). *Information that influences economic decisions because it differs between alternative courses of action.*

Definition 3.3 (Irrelevant Information). *Information that does not affect the outcome of a decision and therefore should not influence economic analysis.*

In everyday contexts, the distinction between relevant and irrelevant information can be subjective. In economics and managerial accounting, however, the distinction is more objective: information is relevant only if it affects the comparison between alternative decisions.

3.2 Cost-Volume-Profit (CVP) Relationships

To simplify CVP calculations, managers typically adopt several assumptions regarding prices, costs, and sales mix.

- **Selling price is constant:** the price of a product or service does not change as the sales volume changes.
- **Costs are linear:** costs can be divided into **variable** and **fixed** components. Variable costs remain constant **per unit**, while fixed costs remain constant **in total** within the relevant range.
- **Constant sales mix:** in multiproduct companies, the proportion in which different products are sold remains constant.

3.2.1 LO1: Changes in Activity, Contribution Margin, and Net Operating Income

Definition 3.4 (Contribution Margin (CM)). *The amount remaining from sales revenue after variable expenses have been deducted.*

The contribution margin is used first to cover **fixed expenses**. Any remaining contribution margin becomes **net operating income** (profit).

Using formulas:

$$\text{Contribution Margin} = \text{Sales} - \text{Variable Expenses}$$

$$\text{Net Operating Income} = \text{Contribution Margin} - \text{Fixed Expenses}$$

Where:

$$\text{Sales} = \text{Selling Price Per Unit } (P) \times \text{Quantity Sold } (Q)$$

$$\text{Variable Expenses} = \text{Variable Expenses Per Unit } (V) \times \text{Quantity Sold } (Q)$$

Therefore:

$$\text{Profit} = (P \times Q - V \times Q) - \text{Fixed Expenses}$$

Factoring Q :

$$\text{Profit} = (P - V) \times Q - \text{Fixed Expenses}$$

Where:

$$\text{Unit Contribution Margin} = P - V$$

Using this definition, the profit equation becomes:

$$\text{Profit} = \text{Unit CM} \times Q - \text{Fixed Expenses}$$

EXAMPLE

Consider the following contribution income statement:

Racing Bicycle Company Contribution Income Statement For the Month of June		
Sales (500 bicycles)	\$	250,000
Less: Variable expenses		150,000
Contribution margin		100,000
Less: Fixed expenses		80,000
Net operating income	\$	20,000

It is often useful to express values on a **per-unit basis** to highlight its contribution toward covering fixed expenses and generating profit:

Racing Bicycle Company Contribution Income Statements For the Month of June		
	Total	For Unit
Sales (500 bicycles)	\$ 250,000	\$ 500
Less: Variable expenses	150,000	300
Contribution margin	100,000	\$ 200
Less: Fixed expenses	80,000	
Net operating income	\$ 20,000	

This means that each additional unit sold contributes 200 \$.

Definition 3.5 (Break-Even Point). *The level of sales at which profit is zero.*

The break-even point is reached when Contribution Margin = Fixed Expenses:

Racing Bicycle Company Contribution Income Statement, For the Month of June		
	Total	Per Unit
Sales (400 bicycles)	\$ 200,000	\$ 500
Less: Variable expenses	120,000	300
Contribution margin	80,000	\$ 200
Less: Fixed expenses	80,000	
Net operating income	\$ -	

In our example, the break-even point is reached when the company sells 400 units:

$$400 \times 200 \$ = 80.000 \$$$

which exactly covers the fixed expenses (Profit = 0 \$).

Racing Bicycle Company Contribution Income Statement For the Month of June		
	Total	Per Unit
Sales (401 bicycles)	\$ 200,500	\$ 500
Less: Variable expenses	120,300	300
Contribution margin	80,200	\$ 200
Less: Fixed expenses	80,000	
Net operating income	\$ 200	

If the company sells one more unit for a total of 401 units, the additional contribution margin is 200 \$. Therefore:

$$\text{Net operating income} = 200 \$$$

We would have gotten the same result using the formulas previously introduced:

$$\text{Profit} = \text{Unit CM} \times Q - \text{Fixed Expenses}$$

$$\text{Profit} = (500 \$ - 300 \$) \times 401 - 80.000 \$ = 200 \$ \times 401 - 80.000 \$ = 80.200 \$ - 80.000 \$ = 200 \$$$

3.2.2 LO2: CVP Graph & Profit Graph

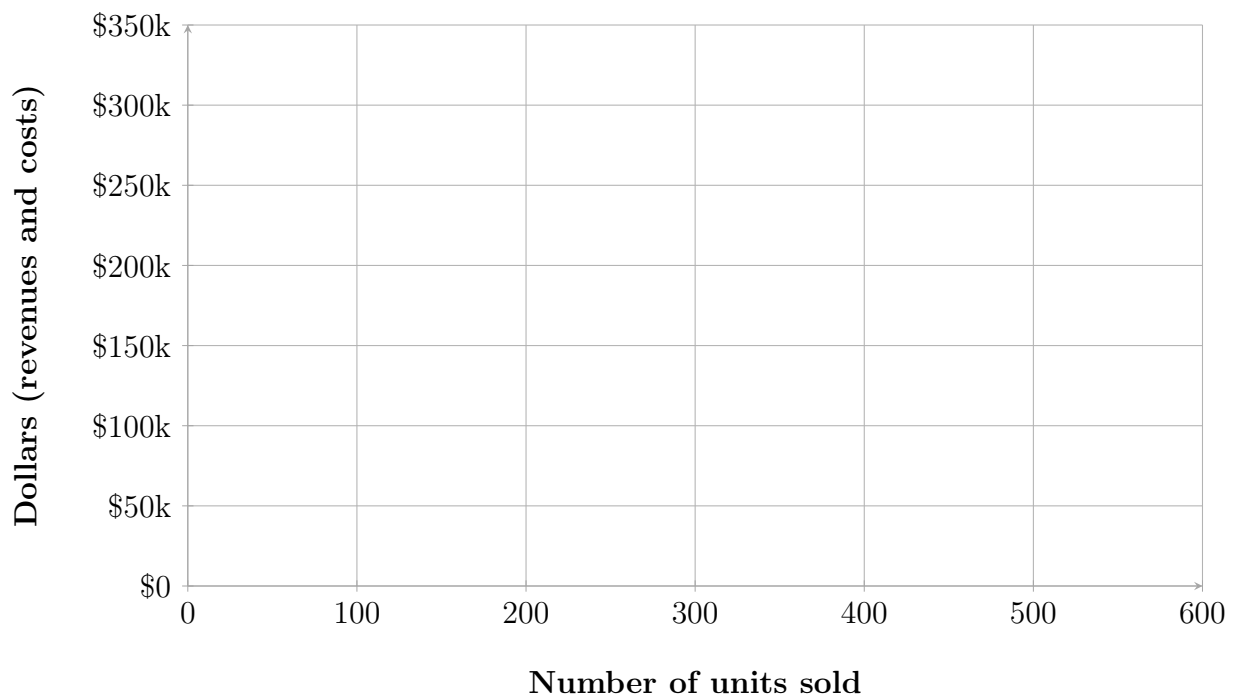
3.2.2.1 CVP Graph

The relationships among revenue, cost, profit, and volume can also be represented graphically using a **CVP graph**. Let's consider the following income statement prepared at 0, 200, 400 and 600 units sold:

	Units Sold			
	0	200	400	600
Sales	\$ -	\$ 100,000	\$ 200,000	\$ 300,000
Total variable expenses	-	60,000	120,000	180,000
Contribution margin	-	40,000	80,000	120,000
Fixed expenses	80,000	80,000	80,000	80,000
Net operating income (loss)	\$ (80,000)	\$ (40,000)	\$ -	\$ 40,000

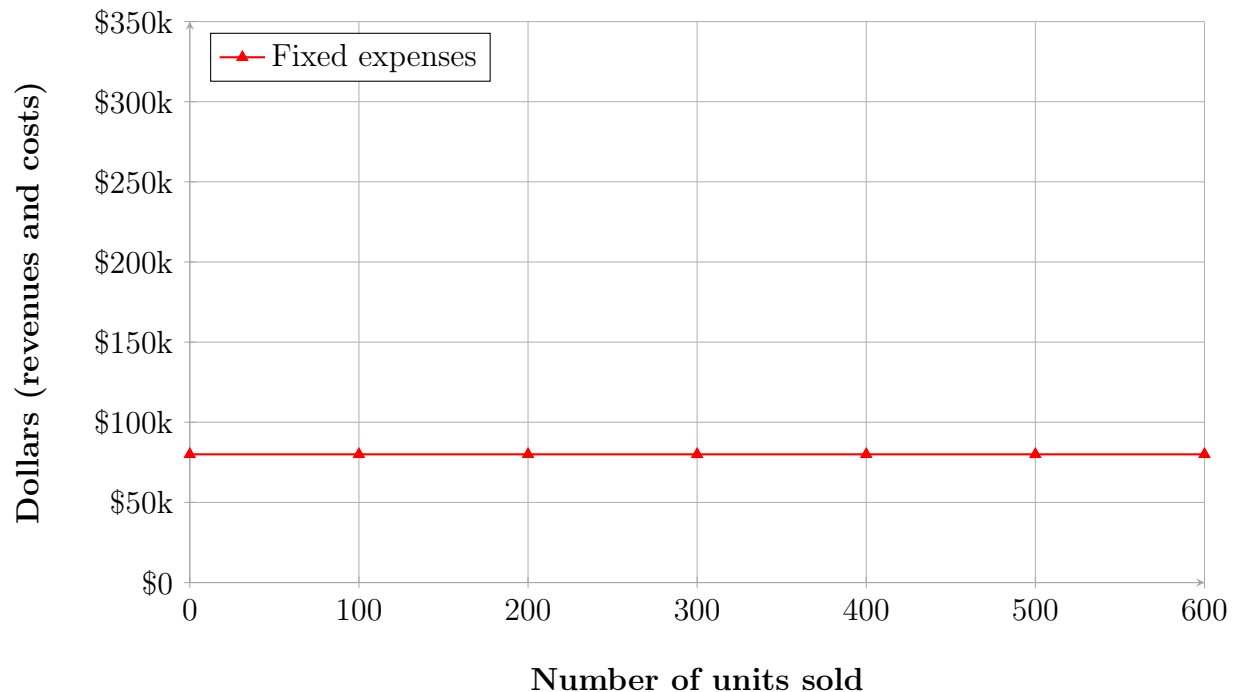
In a CVP graph:

- The **horizontal axis (X)** represents the number of units sold.
- The **vertical axis (Y)** represents dollars (revenues and costs).



Step 1: Plot Fixed Costs

We first draw a horizontal line parallel to the horizontal axis representing total **fixed expenses** (we assume 80.000):



Since fixed costs do not change with the number of units sold, this line remains constant.

Step 2: Plot Total Expenses

We choose a level of activity (we assume 400 units) and compute total expenses:

$$\text{Total expenses} = \text{Fixed expenses} + \text{Variable expenses}$$

If:

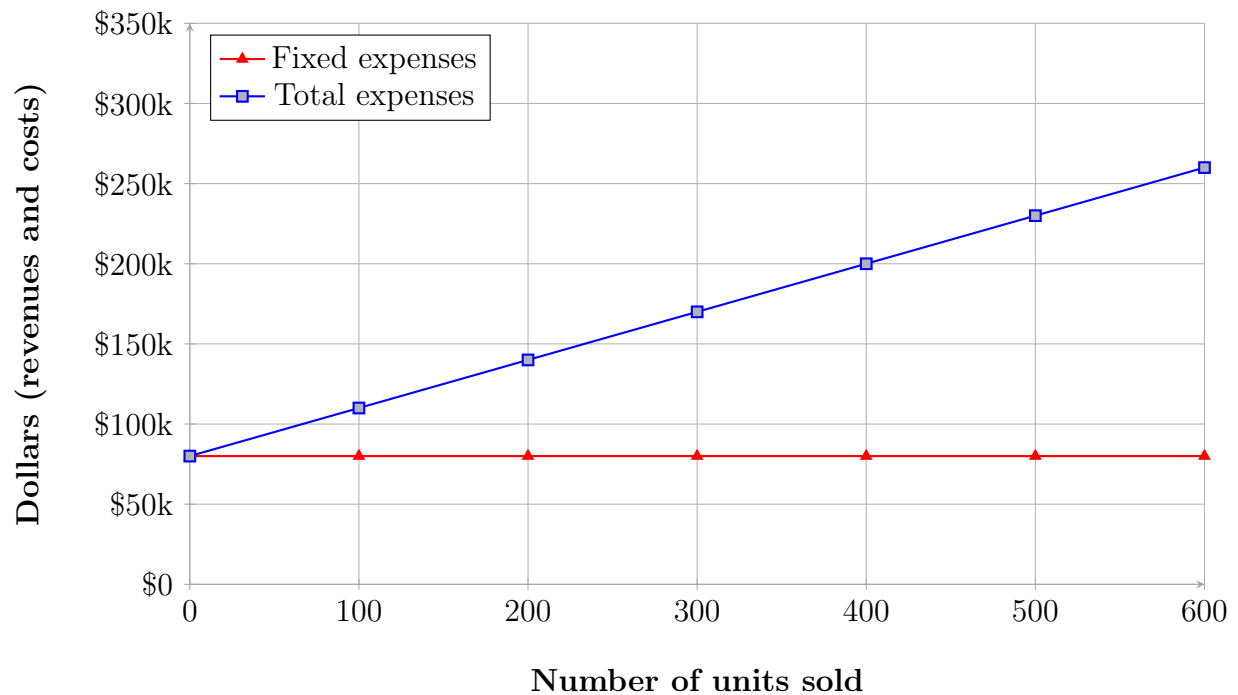
$$\text{Variable cost per unit} = 300$$

then:

$$\text{Variable expenses} = 400 \times 300 = 120.000$$

$$\text{Total expenses} = 80.000 + 120.000 = 200.000$$

So we plot the point and draw the **total expenses line** starting from the fixed-cost intercept:

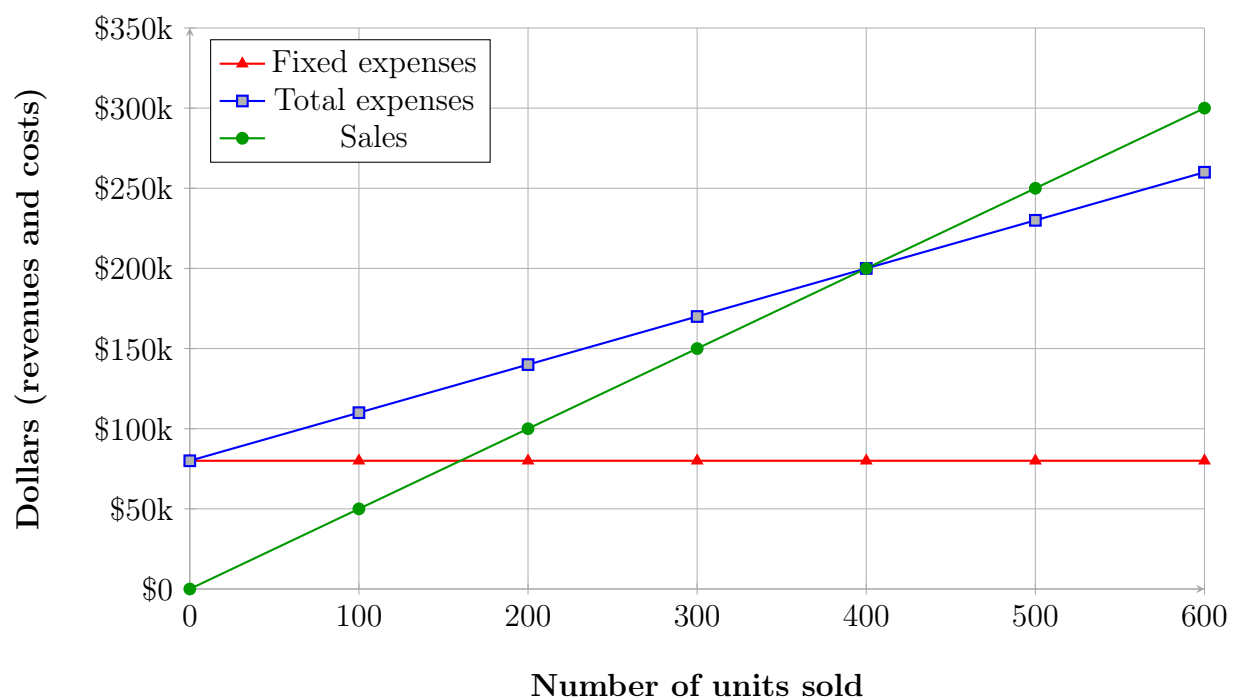


Step 3: Plot the Sales Line

We then compute sales for the same level of activity:

$$\text{Selling price per unit} = 500 \Rightarrow \text{Sales} = 400 \times 500 = 200.000$$

We plot this point and draw the **sales line** starting from the origin:

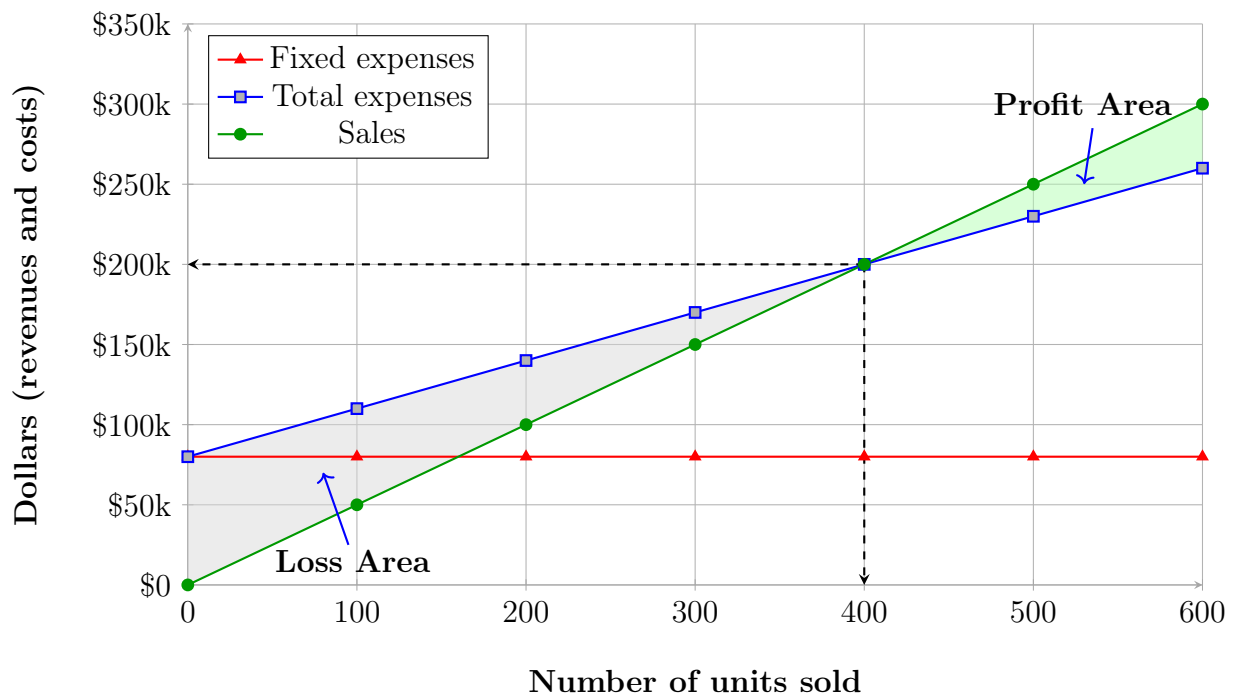


Step 4: Find the Break-Even Point

The **break-even point** occurs where the sales line intersects the total expenses line:

$$\text{Sales} = \text{Total expenses}$$

$$\text{Profit} = 0$$



In this example:

$$\text{Break-even} = 400 \text{ units or } \$200.000 \text{ in sales}$$

We can also see that:

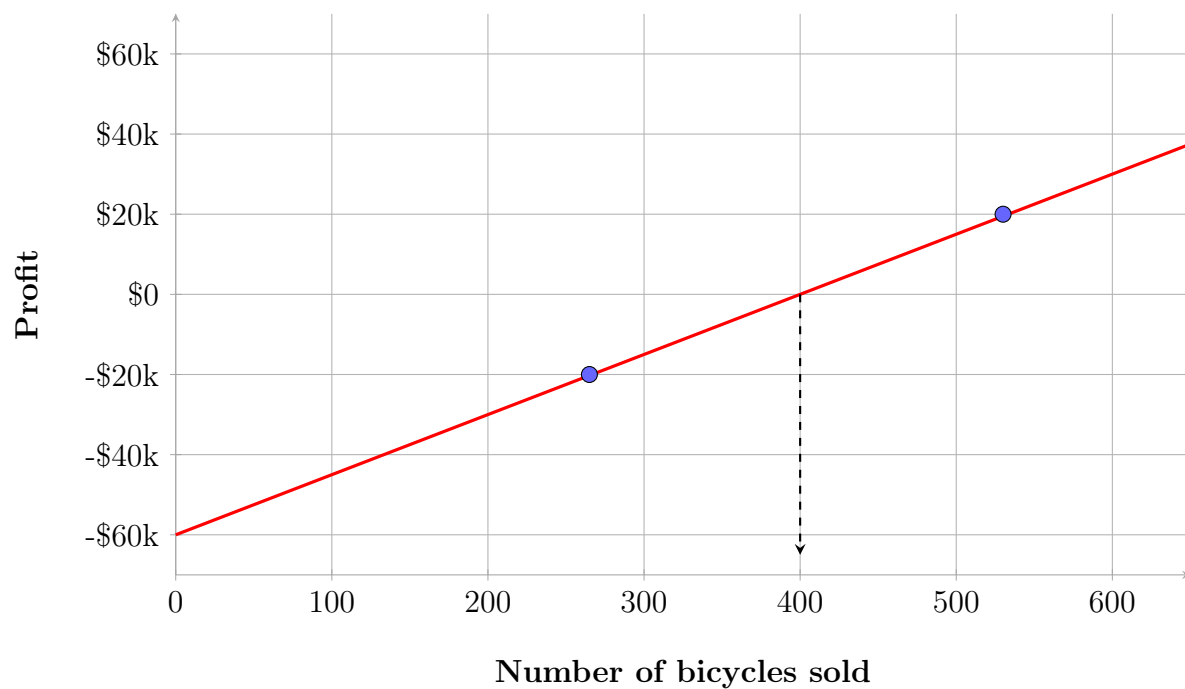
- To the **left of the break-even point**: the company operates at a **loss**.
- To the **right of the break-even point**: the company generates a **profit**.

3.2.2.2 Profit Graph

A simpler representation of CVP relationships is the **profit graph**, which shows the relationship between profit and the number of units sold.

In a profit graph:

- The horizontal axis represents units sold.
- The vertical axis represents profit.



The profit equation is:

$$\text{Profit} = \text{Unit CM} \times Q - \text{Fixed costs}$$

where:

$$\text{Unit CM} = P - V$$

For example:

$$\text{Unit CM} = 500 - 350 = 150$$

$$\text{Profit} = 150Q - 60,000$$

When:

$$Q = 400$$

$$\text{Profit} = 0$$

which corresponds to the **break-even point**.

3.2.3 LO3: Using the CM/VE Ratio

Definition 3.6 (Contribution Margin Ratio). *The contribution margin expressed as a percentage of sales.*

$$\text{CM Ratio} = \frac{\text{Contribution Margin}}{\text{Sales}}$$

Alternatively, the CM ratio can also be computed using per-unit values:

$$\text{CM Ratio} = \frac{\text{Contribution Margin per Unit}}{\text{Selling Price per Unit}}$$

Racing Bicycle Company Contribution Income Statement, For the Month of June		
	Total	Per Unit
Sales (400 bicycles)	\$ 200,000	\$ 500
Less: Variable expenses	120,000	300
Contribution margin	80,000	\$ 200
Less: Fixed expenses	80,000	
Net operating income	\$ -	

$$\text{CM Ratio} = \frac{\$80,000}{\$200,000} = 40\% \quad \text{or} \quad \text{CM Ratio} = \frac{\$200}{\$500} = 40\%$$

The CM ratio indicates how much contribution margin is generated for each dollar of sales:

For each \$1 of sales → +\$0.40 contribution margin

Definition 3.7 (Variable Expense Ratio). *The variable expenses expressed as a percentage of sales.*

$$\text{Variable Expense Ratio} = \frac{\text{Variable Expenses}}{\text{Sales}}$$

Alternatively, the VE ratio can also be computed using per-unit values:

$$\text{VE Ratio} = \frac{\text{Variable Expenses per Unit}}{\text{Selling Price per Unit}}$$

Racing Bicycle Company Contribution Income Statement, For the Month of June		
	Total	Per Unit
Sales (400 bicycles)	\$ 200,000	\$ 500
Less: Variable expenses	120,000	300
Contribution margin	80,000	\$ 200
Less: Fixed expenses	80,000	
Net operating income	\$ -	

$$\text{VE Ratio} = \frac{\$120,000}{\$200,000} = 60\% \quad \text{or} \quad \text{VE Ratio} = \frac{\$300}{\$500} = 60\%$$

The CM Ratio and VE Ratio are mathematically related to one another:

$$\text{CM Ratio} = \frac{\text{Contribution Margin}}{\text{Sales}} = \frac{\text{Sales} - \text{Variable Expenses}}{\text{Sales}} = 1 - \text{VE Ratio}$$

$$\text{CM Ratio} = 1 - \text{VE Ratio} = 1 - 0.60 = 0.40 = 40\%$$

The change in contribution margin due to a change in sales can be computed as:

$$\Delta \text{CM} = \text{CM Ratio} \times \Delta \text{Sales}$$

For example, if sales increase by \$50,000:

Racing Bicycle Company Contribution Income Statements For the Month of June		
	Total	For Unit
Sales (500 bicycles)	\$ 250,000	\$ 500
Less: Variable expenses	150,000	300
Contribution margin	100,000	\$ 200
Less: Fixed expenses	80,000	
Net operating income	\$ 20,000	

$$\Delta CM = 0.40 \times \$50,000 = \$20,000$$

Since fixed costs remain unchanged:

$$\Delta \text{Profit} = \Delta CM = \$20,000$$

The profit equation:

$$\text{Profit} = \text{Unit CM} \times Q - \text{Fixed Expenses}$$

can also be expressed in terms of CM Ratio (%) and sales (\$):

$$\text{Profit} = \text{CM Ratio} \times \text{Sales} - \text{Fixed Expenses}$$

$$\text{Profit} = 0.40 \times \$250,000 - \$80,000 = \$20,000$$

3.2.4 LO4: Effects on Net Operating Income

The net operating income depends on four main factors:

- Selling price
- Variable costs
- Fixed costs
- Sales volume

Any change in one or more of these variables will affect profit. To analyze changes in profit, it is sufficient to focus only on the variations (**incremental analysis**), instead of recomputing the entire income statement.

Case 1: Change in Fixed Costs and Volume

- Increasing fixed costs increases the break-even point
- Increasing volume increases contribution margin

$$\Delta \text{Profit} = \Delta CM - \Delta F$$

Or similarly, but only if variable expenses are **constant**:

$$\Delta \text{Profit} = (\Delta \text{Sales} \times CM \text{ Ratio}) - \Delta F = (\Delta Q \times \text{Unit CM}) - \Delta F$$

For example:

	500 units	540 units
Sales	\$ 250,000	\$ 270,000
Less: Variable expenses	150,000	162,000
Contribution margin	100,000	108,000
Less: Fixed expenses	80,000	90,000
Net operating income	\$ 20,000	\$ 18,000

- Sales increase from 500 to 540 units
- Fixed costs increase by \$10,000 (advertising)

$$\Delta CM = 0.40 \times \$20,000 = 40 \times \$200 = \$8,000$$

$$\Delta \text{Profit} = \$8,000 - \$10,000 = -\$2,000$$

Observation 3.2. *An increase in fixed costs can more than offset the additional contribution margin generated by higher sales, leading to a decrease in profit.*

Case 2: Change in Variable Cost and Volume

- Increasing variable cost reduces the contribution margin
- Higher volume may compensate this reduction

For example:

	500 units	580 units
Sales	\$ 250,000	\$ 290,000
Less: Variable expenses	150,000	179,800
Contribution margin	100,000	110,200
Less: Fixed expenses	80,000	80,000
Net operating income	\$ 20,000	\$ 30,200

- Sales increase from 500 to 580 units
- Variable cost per unit increases from \$300 to \$310
- Contribution margin per unit decreases from \$200 to \$190

Since fixed costs do not change:

$$\Delta \text{Profit} = \Delta \text{CM} = \$10,200$$

Observation 3.3. *An increase in variable cost reduces the contribution margin per unit, but a sufficiently large increase in sales volume can still lead to higher profit.*

Case 3: Change in Selling Price, Fixed Costs, and Volume

- Decreasing price reduces CM
- Increasing fixed costs reduces profit
- Increasing volume increases total CM

For example:

	500 units	650 units
Sales	\$ 250,000	\$ 312,000
Less: Variable expenses	150,000	195,000
Contribution margin	100,000	117,000
Less: Fixed expenses	80,000	95,000
Net operating income	\$ 20,000	\$ 22,000

- Sales increase from 500 to 650 units
- Selling price decreases from \$500 to \$480
- Contribution margin per unit decreases from \$200 to \$180
- Fixed costs increase by \$15,000

$$\Delta \text{Profit} = \Delta \text{CM} - \Delta \text{F} = \$17,000 - \$15,000 = \$2,000$$

Observation 3.4. *When multiple factors change simultaneously, the effect on profit depends on the balance between an increase in sales volume, a decrease in contribution margin and an increase in fixed costs.*

Case 4: Shift from Fixed to Variable Costs

- Fixed costs decrease
- Variable costs increase
- Risk structure changes:
 - Lower fixed costs → lower risk
 - Higher variable costs → lower contribution margin per unit

For example:

	500 units	575 units
Sales	\$ 250,000	\$ 287,500
Less: Variable expenses	150,000	181,125
Contribution margin	100,000	106,375
Less: Fixed expenses	80,000	74,000
Net operating income	\$ 20,000	\$ 32,375

- Sales increase from 500 to 575 units
- Fixed costs decrease (removal of salaries) by \$6.000
- Variable cost per unit increases (sales commission) from \$300 to \$315
- Contribution margin per unit decreases from \$200 to \$185

$$\Delta \text{Profit} = \Delta \text{CM} - \Delta \text{F} = \$6.375 - (-\$6.000) = \$12.375$$

Observation 3.5. *Shifting from fixed costs to variable costs reduces fixed risk but also reduces the contribution margin per unit.*

Case 5: Change in Selling Price (Special Order)

- Target increase in profit
- Additional units to sell

For example:

- 150 units to sell (special order)
- Target profit: \$3.000
- Variable cost per unit: \$300

- Fixed costs remain constant

In order to reach the target increase in profit, the firm must increase the price of the product:

$$\text{Required CM per unit} = \frac{\text{Target Profit}}{\Delta Q} = \frac{\$3.000}{150} = \$20$$

$$\text{Required selling price} = \text{Variable Cost per Unit} + \text{Required CM per Unit} = \$300 + \$20 = \$320$$

In fact:

$$\text{Sales} = P \times Q = \$320 \times 150 = \$48.000$$

$$\text{Variable Expenses} = \text{Variable Cost per Unit} \times Q = \$300 \times 150 = \$45.000$$

And since fixed costs do not change:

$$\Delta \text{Profit} = \Delta \text{CM} = \text{Required CM per unit} \times Q = \$20 \times 150 = \$3.000$$

Observation 3.6. *When fixed costs do not change, the required selling price must cover the variable cost per unit and the desired contribution margin per unit.*

3.2.5 LO5: Determining the Break-Even Point

Break-even analysis can be used to determine:

- the number of units that must be sold to earn zero profit,
- the dollar sales required to earn zero profit.

Break-even analysis shows the minimum sales level required to avoid a loss.

Break-Even in Unit Sales

For a single-product company, the break-even point can be found using the basic profit equation:

$$\text{Profit} = \text{Unit CM} \times Q - \text{Fixed expenses}$$

At break-even:

$$\text{Profit} = 0$$

Therefore:

$$0 = \text{Unit CM} \times Q - \text{Fixed expenses}$$

For example:

Racing Bicycle Company Contribution Income Statement For the Month of June			
	Total	Per Unit	CM Ratio
Sales (500 bicycles)	\$ 250,000	\$ 500	100%
Less: Variable expenses	150,000	300	60%
Contribution margin	100,000	\$ 200	40%
Less: Fixed expenses	80,000		
Net operating income	\$ 20,000		

$$0 = \$200 \times Q - \$80.000$$

$$200Q = \$80.000$$

$$Q = \frac{\$80.000}{\$200} = 400$$

Thus, the break-even point in unit sales is 400.

Alternatively, we can use the formula:

$$\text{Break-Even Unit Sales} = \frac{\text{Fixed Expenses}}{\text{Unit CM}} = \frac{\$80.000}{\$200} = 400$$

Break-Even in Dollar Sales

Break-even can also be expressed in terms of sales revenue.

Using the equation method:

$$\text{Profit} = \text{CM Ratio} \times \text{Sales} - \text{Fixed Expenses}$$

At break-even:

$$0 = \text{CM Ratio} \times \text{Sales} - \text{Fixed Expenses}$$

For example:

Racing Bicycle Company Contribution Income Statement For the Month of June			
	Total	Per Unit	CM Ratio
Sales (500 bicycles)	\$ 250,000	\$ 500	100%
Less: Variable expenses	150,000	300	60%
Contribution margin	100,000	\$ 200	40%
Less: Fixed expenses	80,000		
Net operating income	\$ 20,000		

$$0 = 0.40 \times \text{Sales} - \$80,000$$

$$0.40 \times \text{Sales} = \$80,000$$

$$\text{Sales} = \frac{\$80,000}{0.40} = \$200,000$$

Thus, the break-even point in dollar sales is \$200,000.

Alternatively, we can use the formula:

$$\text{Break-Even Dollar Sales} = \frac{\text{Fixed Expenses}}{\text{CM Ratio}} = \frac{\$80,000}{0.40} = \$200,000$$

In general, in our example:

- If sales are below 400 units, the company incurs a loss.
- If sales are exactly 400 units, profit is zero.
- If sales exceed 400 units, the company earns a profit.

3.2.6 LO6: Target Profit Analysis

Definition 3.8 (Target Profit Analysis). *An analysis that determines the level of sales needed to achieve a desired profit.*

Target Profit in Unit Sales

$$\text{Profit} = \text{Unit CM} \times Q - \text{Fixed Expenses}$$

Solve for Q :

$$Q = \frac{\text{Target profit} + \text{Fixed Expenses}}{\text{Unit CM}}$$

For example, suppose a certain firm wants to know how many units must be sold to earn a target profit of \$100,000:

$$Q = \frac{\$100,000 + \$80,000}{\$200} = 900 \text{ Units}$$

Alternatively, we can use the formula:

$$\text{Unit Sales for Target Profit} = \frac{\text{Target Profit} + \text{Fixed Expenses}}{\text{Unit CM}} = 900$$

Target Profit in Dollar Sales

$$\text{Profit} = \text{CM ratio} \times \text{Sales} - \text{Fixed Expenses}$$

Solve for Sales:

$$\text{Sales} = \frac{\text{Target profit} + \text{Fixed expenses}}{\text{CM ratio}}$$

For example, suppose a certain firm wants to know the sales volume that must be generated to earn a target profit of \$100,000:

$$\text{Sales} = \frac{\$100,000 + \$80,000}{0.40} = \$450,000$$

Alternatively, we can use the formula:

$$\text{Dollar sales for Target Profit} = \frac{\text{Target Profit} + \text{Fixed Expenses}}{\text{CM ratio}} = \$450,000$$

3.2.7 LO7: Margin of Safety and Cost Structure

Definition 3.9 (Margin of Safety). *The excess of actual (or budgeted) sales over break-even sales. It represents how much sales can drop before the company incurs a loss.*

Observation 3.7. *A higher margin of safety indicates lower risk, since sales can decline more before reaching the break-even point.*

Definition 3.10 (Cost Structure). *The relative proportion of fixed and variable costs in a company.*

Margin of Safety in Dollars

$$\text{Margin of Safety} = \text{Total Sales} - \text{Break-Even Sales}$$

For example:

	Break-even sales 400 units	Actual sales 500 units
Sales	\$ 200,000	\$ 250,000
Less: variable expenses	120,000	150,000
Contribution margin	80,000	100,000
Less: fixed expenses	80,000	80,000
Net operating income	\$ -	\$ 20,000

$$\text{Margin of safety} = \$250.000 - \$200.000 = \$50.000$$

Margin of Safety Percentage

$$\text{Margin of safety \%} = \frac{\text{Margin of Safety}}{\text{Total Sales}}$$

In reference to the previous example:

$$\text{Margin of safety \%} = \frac{\$50.000}{\$250.000} = 20\%$$

Margin of Safety in Units

$$\text{Margin of Safety (units)} = \frac{\text{Margin of Safety (dollars)}}{\text{Selling Price per Unit}}$$

In reference to the previous example:

$$\text{Margin of Safety (units)} = \frac{\$50.000}{\$500} = 100 \text{ units}$$

High vs Low Fixed Cost Structures

- High fixed costs (low variable costs):
 - Higher profits in good years

- Lower profits (or losses) in bad years
- Low fixed costs (high variable costs):
 - Lower profits in good years
 - More stable income across different levels of sales

Observation 3.8. *Companies with lower fixed costs have less risk and more stable income, while companies with higher fixed costs have higher risk but greater profit potential.*

3.2.8 LO8: Operating Leverage

Definition 3.11 (Operating Leverage). *Measure of how sensitive net operating income is to percentage changes in sales.*

$$\text{Degree of Operating Leverage (DOL)} = \frac{\text{Contribution Margin}}{\text{Net Operating Income}}$$

For example:

	Actual sales 500 Bikes
Sales	\$ 250,000
Less: variable expenses	150,000
Contribution margin	100,000
Less: fixed expenses	80,000
Net income	\$ 20,000

$$\text{DOL} = \frac{\$100,000}{\$20,000} = 5$$

Observation 3.9. *A higher degree of operating leverage means that a small percentage change in sales results in a larger percentage change in profit.*

$$\%\Delta\text{Profit} = \text{DOL} \times \%\Delta\text{Sales}$$

In reference to the previous example:

$$\%\Delta\text{Profit} = 5 \times 10\% = 50\%$$

This means that if the firm increases its sales by 10%, net operating income will increase by 50%. We can verify this:

	Actual sales (500)	Increased sales (550)
Sales	\$ 250,000	\$ 275,000
Less variable expenses	150,000	165,000
Contribution margin	100,000	110,000
Less fixed expenses	80,000	80,000
Net operating income	\$ 20,000	\$ 30,000

Using the percentage change formula:

$$\% \Delta \text{Profit} = \frac{\text{New Value} - \text{Old Value}}{\text{Old Value}} = \frac{\$30,000 - \$20,000}{\$20,000} = 50\%$$

Observation 3.10. *Operating leverage is driven by cost structure:*

- *High fixed costs $\Rightarrow$ high operating leverage $\Rightarrow$ higher risk and higher potential returns*
- *Low fixed costs $\Rightarrow$ low operating leverage $\Rightarrow$ lower risk and more stable profits*

Sales Commissions and Incentives

Sales commissions can be based on:

- Selling price (sales revenue)
- Contribution margin

For example, consider a firm that produces two types of surfboards:

- XR7:

$$\text{Price} = \$100, \quad \text{CM per unit} = \$25$$

- Turbo:

$$\text{Price} = \$150, \quad \text{CM per unit} = \$18$$

Observation 3.11 (Bad Approach). *If commissions are based on sales revenue, sales-people may prefer products with higher prices rather than higher contribution margins.*

Observation 3.12 (Correct Approach). *Aligning incentives with contribution margin ensures that sales decisions maximize company profit.*

3.2.9 LO9: Multiproduct Break-Even and Sales Mix

Definition 3.12 (Sales Mix). *The relative proportion in which a company's different products are sold.*

In a multiproduct company, break-even analysis is more complex because different products have different:

- selling prices
- variable cost structures
- contribution margins

For break-even analysis, it is usually assumed that the sales mix remains constant.

For example, suppose a firm sells two products with the following sales mix:

- bicycles (45%)
- carts (55%)

	Bicycle		Carts		Total	
Sales	\$ 250,000	100%	\$ 300,000	100%	\$ 550,000	100.0%
Variable expenses	150,000	60%	135,000	45%	285,000	51.8%
Contribution margin	<u>100,000</u>	<u>40.0%</u>	<u>165,000</u>	<u>55%</u>	<u>265,000</u>	<u>48.2%</u>
Fixed expenses					170,000	
Net operating income					<u>\$ 95,000</u>	
Sales mix	\$ 250,000	45%	\$ 300,000	55%	\$ 550,000	100%

The total contribution margin ratio is computed using total values:

$$\text{CM ratio} = \frac{\$265,000}{\$550,000} \approx 48.2\%$$

The break-even formula in dollar sales is still:

$$\text{Break-Even Dollar Sales} = \frac{\text{Fixed expenses}}{\text{CM ratio}}$$

Using the multiproduct data:

$$\text{Break-Even Dollar Sales} = \frac{\$170,000}{48.2\%} = \$352,697$$

This means that if the sales mix remains constant at 45% bicycles and 55% carts, the firm must generate approximately \$352.697 in total sales revenue to break even. These break-even sales are then allocated according to the sales mix:

$$\text{Bicycles} = 45\% \times \$352.697 \approx \$158.714$$

$$\text{Carts} = 55\% \times \$352.697 \approx \$193.983$$

Observation 3.13. *A shift in the sales mix affects the overall CM ratio:*

- *more sales of high-CM products $\Rightarrow$ higher overall CM ratio $\Rightarrow$ lower break-even point*
- *more sales of low-CM products $\Rightarrow$ lower overall CM ratio $\Rightarrow$ higher break-even point*

3.3 Quick Checks

1. Coffee Klatch is an espresso stand in a downtown office building. The average selling price of a cup of coffee is \$1.49 and the average variable expense per cup is \$0.36. The average fixed expense per month is \$1.300. An average of 2.100 cups are sold each month. What is the CM ratio for Coffee Klatch?

- A. 1.319
- B. 0.758
- C. 0.242
- D. 4.139

Correct answer(s): B.

Explanation:

$$\text{CM Ratio} = \frac{\$1.49 - \$0.36}{\$1.49} = \frac{\$1.13}{\$1.49} = 0.758$$

2. Coffee Klatch is an espresso stand in a downtown office building. The average selling price of a cup of coffee is \$1.49 and the average variable expense per cup is \$0.36. The average fixed expense per month is \$1.300. An average of 2.100 cups are sold each month. What is the break-even sales in dollars?

- A. \$1.300
- B. \$1.715
- C. \$1.788
- D. \$3.129

Correct answer(s): B.

Explanation:

$$\text{Break-Even Dollar Sales} = \frac{\$1.300}{0.758} = \$1.715$$

3. Coffee Klatch is an espresso stand in a downtown office building. The average selling price of a cup of coffee is \$1.49 and the average variable expense per cup is \$0.36. The average fixed expense per month is \$1.300. An average of 2.100 cups are sold each month. What is the break-even point in units?

- A. 872 cups
- B. 3.611 cups
- C. 1.200 cups
- D. 1.150 cups

Correct answer(s): D.

Explanation:

$$\text{Break-Even Unit Sales} = \frac{\$1.300}{\$1.13} = 1.150 \text{ cups}$$

4. Coffee Klatch is an espresso stand in a downtown office building. The average selling price of a cup of coffee is \$1.49 and the average variable expense per cup is \$0.36. The average fixed expense per month is \$1.300. Use the formula method to determine how many cups must be sold to attain a target profit of \$2.500.

- A. 3.363 cups
- B. 2.212 cups
- C. 1.150 cups
- D. 4.200 cups

Correct answer(s): A.

Explanation:

$$Q = \frac{\$2.500 + \$1.300}{\$1.13} = \frac{\$3.800}{\$1.13} = 3.363$$

5. Coffee Klatch is an espresso stand in a downtown office building. The average selling price of a cup of coffee is \$1.49 and the average variable expense per cup is \$0.36. The average fixed expense per month is \$1,300. Use the formula method to determine the sales dollars required to attain a target profit of \$2,500.

- A. \$2,550
- B. \$5,013
- C. \$8,458
- D. \$10,555

Correct answer(s): B.

Explanation:

$$\text{Dollar Sales} = \frac{\$3,800}{0.758} = \$5,013$$

6. Coffee Klatch is an espresso stand in a downtown office building. The average selling price of a cup of coffee is \$1.49 and the average variable expense per cup is \$0.36. The average fixed expense per month is \$1,300. An average of 2,100 cups are sold each month. What is the margin of safety expressed in cups?

- A. 3,250 cups
- B. 950 cups
- C. 1,150 cups
- D. 2,100 cups

Correct answer(s): B.

Explanation:

$$\text{Margin of safety in units} = 2,100 - 1,150 = 950 \text{ cups}$$

7. Coffee Klatch is an espresso stand in a downtown office building. The average selling price of a cup of coffee is \$1.49 and the average variable expense per cup is \$0.36. The average fixed expense per month is \$1,300. An average of 2,100 cups are sold each month. What is the degree of operating leverage?

- A. 2.21
- B. 0.45
- C. 0.34
- D. 2.92

Correct answer(s): A.

Explanation:

$$\text{Degree of operating leverage} = \frac{\$2.373}{\$1.073} = 2.21$$

8. At Coffee Klatch the average selling price of a cup of coffee is \$1.49, the average variable expense per cup is \$0.36, the average fixed expense per month is \$1,300, and an average of 2,100 cups are sold each month. If sales increase by 20%, by how much should net operating income increase?

- A. 30.0%
- B. 20.0%
- C. 22.1%
- D. 44.2%

Correct answer(s): D.

Explanation:

$$\% \Delta \text{Net operating income} = 2.21 \times 20\% = 44.2\%$$